LECTURE L3-4 · MODULE 1 · CO1

Data types and representation

Week 2 | Slides: `DATA209_Module_1_Statistical_and_Data_Foundations.pptx`, pages 20–30 | Laboratory: **P3-4**

TOPICS COVERED

- Nominal, ordinal, interval and ratio scales
- Discrete vs. continuous
- Structured vs. semi-structured vs. unstructured data
- Text representation basics
- Time-series behaviour: trend and seasonality

Measurement scale decides the statistic

The dtype pandas infers is a storage decision. The **measurement scale** is a semantic one, and it governs which operations are meaningful.

SCALE	PROPERTY	LEGAL	ILLEGAL
Nominal	names, no order	counts, mode, chi-square	mean, median, ordering
Ordinal	ordered, unequal gaps	median, percentiles, rank tests	mean (debatable), differences
Interval	equal gaps, arbitrary zero	mean, standard deviation, differences	ratios
Ratio	equal gaps, true zero	everything, including "twice as much"	—

An integer column can be nominal. In the shoppers dataset `OperatingSystems`, `Browser`, `Region` and `TrafficType` are stored as integers and are pure codes; averaging them produces a number with no referent. This is the single most common error in submitted assignments.

Independently of scale, variables are **discrete** (countable — page views, pregnancies) or **continuous** (any value in a range — BMI, duration). Discrete versus continuous decides the plot; the measurement scale decides the statistic.

Structure

- **Structured** — fixed schema, rows and columns. CSV, SQL, spreadsheets. Directly analysable.
- **Semi-structured** — self-describing but irregular. JSON, XML, logs. Needs flattening first.
- **Unstructured** — no tabular form. Text, images, audio, video. Requires a representation

decision before any analysis at all.

Storage format matters too: CSV carries no dtypes and breaks on delimiters inside text; Excel silently coerces types and hides rows; Parquet is typed and columnar but not human-readable.

Text becomes countable

Raw text is human-readable and machine-useless. Exploration begins after a representation choice:

1. **Tokenise** — split into words; normalise case, strip punctuation, optionally drop stop words.
2. **Bag-of-words** — count each vocabulary term per document. Interpretable; ignores word order.
3. **TF-IDF** — down-weight terms common across all documents, up-weight distinctive ones.
4. **N-grams** capture short phrases; **embeddings** capture meaning — both cost interpretability.

For exploratory purposes, document-length distribution, vocabulary size and ranked term frequency are usually sufficient. A ranked frequency bar chart is precise; a word cloud is decorative.

Time series

Four components are present in every series: **trend** (long-run direction), **seasonality** (a repeating pattern of fixed period), **cyclic** behaviour (long waves of no fixed period) and the **irregular** residual.

Two rules follow, and both matter for the final project: with time-ordered data you may **not shuffle rows**, and you may **not split randomly**. Training on rows recorded after the test rows is temporal leakage, covered properly in L29-30.

An ordered `Categorical` for a month column is not cosmetic. Sorted alphabetically, a seasonal pattern becomes invisible; sorted chronologically, the same data shows it immediately.

KEY TERMS

`Nominal` `Ordinal` `Interval` `Ratio` `Discrete` `Continuous` `Semi-structured data` `Tokenisation` `Bag-of-words` `TF-IDF` `Trend` `Seasonality`

COMMON ERRORS TO AVOID

- Averaging an integer column that encodes a nominal category
- Leaving a month or size column unordered, so plots sort alphabetically
- Counting categories before trimming whitespace and unifying case
- Splitting time-ordered data randomly

READINGS — ALL FREE TO ACCESS

1. pandas user guide — categorical data
   `https://pandas.pydata.org/docs/user_guide/categorical.html`
2. pandas user guide — time series
   `https://pandas.pydata.org/docs/user_guide/timeseries.html`
3. McKinney — Python for Data Analysis, 3rd ed.
   `https://wesmckinney.com/book/`
4. Kaggle Learn — Pandas
   `https://www.kaggle.com/learn/pandas`

TEST YOURSELF

1. Postal codes are integers. What scale are they, and what may you compute?
2. Why is the mean of a five-point Likert scale contested?
3. Name a variable that is discrete and ratio-scaled.
4. Give one reason a random train/test split invalidates a time-series analysis.